

Automated Quantification of Senescent Cells

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Abstract—Senescent cells are commonly identified through β -galactosidase staining and counted manually, a process that is time-consuming and prone to operator variability. We present an automated image-processing pipeline that quantifies senescent cell frequency using DAPI-stained nuclei and β -gal activity extracted from confocal microscopy. Nuclei are segmented using morphological enhancement and distance-transform watershed, while senescence is estimated through blue-channel dominance thresholding. Validation against human annotation ($N = 23$) revealed systematic underestimation with moderate deviation, indicating that β -gal segmentation performs more reliably than nuclei segmentation, and that errors in dapi_n directly impact senescence percentage. This framework provides an extensible foundation for scalable aging-biology quantification using open computer vision methods.

Index Terms—Cellular senescence, microscopy, image segmentation, watershed, automation, β -galactosidase.

I. INTRODUCTION

Senescence is a runtime state where the cell enters an infinite loop. The cell stops dividing, but it refuses to crash or terminate, in a "zombie state". These zombie cells start broadcasting inflammatory signals to neighboring cells, causing the tissue to lag or degrade over time (aging) [1].

A widely used biomarker of senescence is lysosomal β -galactosidase (SA- β -gal) activity, detectable through chromogenic staining. Traditional quantification relies on eye-manual inspection, which is labor-non-scalable, and subjective. The automation of microscopy analysis offers an objective alternative but remains challenged by noise, overlapping nuclei, cell clustering, staining heterogeneity, and illumination variation.

This work proposes a reproducible pipeline combining color segmentation and nuclei separation to estimate senescent cell proportion, with performance validated against human scoring.

II. METHODS

A. Dataset

Rat neuron culture images were acquired using standard DAPI nuclear staining and β -gal senescence assays at different time of *in vitro* incubation measured by days (Days *in Vitro* or DIV). A total of 23 fields of view were manually annotated to create validation data for comparison.

B. Pipeline Overview

Given a pair of images per field of view (DAPI and β -gal), our pipeline performs:

- 1) **Preprocessing:** DAPI images are preprocessed with CLAHE on the L channel in LAB (clipLimit=4.0, tileGridSize=9 × 9), then optional sharpening (3 × 3 kernel), and morphological open (top-hat-like clean-up with 9 × 9 structuring element) is performed. Afterwards we apply Otsu thresholding on grayscale and small-structure removal.
- 2) **Nuclei delineation:** External contours are classified by area; single nuclei are those with area in [min_area_nuc, max_area_nuc], agglomerates are above the max threshold.
- 3) **Senescence mask:** β -gal images are converted into HSV color space, where blue color is segmented according to a preset hue windowing ($H \in [40, 140]$, $S \geq 10$, $V \geq 15$) intersected with a relative blue-dominance mask $D_{rel} = \frac{\max(0, B - \frac{R+G}{2})}{V+\epsilon} \cdot 255 > \text{blue_dom_thresh}$, then morphological open (2x) and close (1x) using 9 × 9 elliptical kernel is applied.
- 4) **Counting senescent cells:** A nucleus is senescent if its centroid lies within a blue region; if none lies inside, count the single nearest nucleus within near_radius. Blue regions strongly overlapped by agglomerates are excluded when ignore_aggl is true; otherwise, each such region counts as one senescent cell and increments total nuclei by one.
- 5) **Senescence percentage:**
$$\text{Senescence\%} = \frac{\# \text{ senescent cells}}{\# \text{ total (single) nuclei}} \times 100.$$

C. Counting criteria

- All nuclei objects larger than 40 pixels and smaller than max_area_nuc were considered single nuclei, therefore included in the total cell count. Above that threshold it was considered as an agglomeration of 2 or more nuclei and this area excluded from the total count of cells.
- Blue zones larger than the selected min_blue_area (usually around 200 to 300 pixels) indicate a senescent mark of all the nuclei with their centroid laying inside that zone or the closest nucleus at most 40 pixels of vicinity indicated by the near_radium parameter.

- If `ignore_aggl` was set to `True`, it excludes all counts of β -gal activity that are completely covered by an agglomeration area.
- Blue zones partially covered with nuclei agglomeration can still be counted if at least half (or `agg_tol`) of the blue zone's area is not overlapped by any agglomeration area.
- If `ignore_aggl` is set to `False`, every overlapped blue zone is counted as one β -gal cell and another count is included in the total cell count.

D. Hyper-Parameters

Multiple hyperparameters were used to achieve the desired detection of nuclei (using `count_dapi_nuclei` function) and β -gal activity around nuclei (using `find_blue_zones`). Finally, counting β -gal activating cells against total cells was performed by the `count_bgal_nuclei` according with the counting criteria mentioned above. A description of every parameter is stated in Table I.

TABLE I
DESCRIPTION OF HYPER-PARAMETERS

Function	Parameter	Details
<code>count_dapi_nuclei</code>	<code>clahe</code>	CLAHE clipLimit=4.0, tileGridSize=(9,9) on L channel
<code>count_dapi_nuclei</code>	<code>sharpen</code>	Applied before tophat
<code>count_dapi_nuclei</code>	<code>tophat</code>	Open with 9x9 kernel
<code>count_dapi_nuclei</code>	<code>low_intensity_tresh</code>	Threshold + OTSU on gray
<code>count_dapi_nuclei</code>	<code>min_area_nuc</code>	Passed from process_chunk default
<code>count_dapi_nuclei</code>	<code>max_area_nuc</code>	Passed from process_chunk default
<code>count_dapi_nuclei</code>	<code>kernel</code>	Sharpen kernel used in call
<code>find_blue_zones</code>	<code>sharpen</code>	Use Sharpen filter
<code>find_blue_zones</code>	<code>kernel</code>	Sharpen kernel used in call
<code>find_blue_zones</code>	<code>min_blue_area</code>	min_blue_area from process_chunk default
<code>find_blue_zones</code>	<code>hue_min</code>	HSV range
<code>find_blue_zones</code>	<code>hue_max</code>	HSV range
<code>find_blue_zones</code>	<code>sat_min</code>	HSV range
<code>find_blue_zones</code>	<code>val_min</code>	HSV range
<code>find_blue_zones</code>	<code>blue_dom_thresh</code>	Relative blue dominance threshold
<code>find_blue_zones</code>	<code>morph</code>	Open(9x9) × 2 + Close(9x9) × 1
<code>count_bgal_nuclei</code>	<code>near_radius</code>	From process_chunk default
<code>count_bgal_nuclei</code>	<code>ignore_aggl</code>	Ignore zones with agglomerations while counting
<code>count_bgal_nuclei</code>	<code>agg_tol</code>	Overlap tolerance with agglomerates

Given that each day of analysis produced a similar image variation, it was decided to split the hyperparameters to perform the analysis according to the batch it belongs. The only parameters that varied across batches were `max_area_nuc`, `min_blue_area`, `near_radius`, and `ignore_aggl` while the rest were fixed across conditions. Specific hyperparameters are displayed in Table II

TABLE II
LIST OF HYPER-PARAMETERS FOR EACH PROCESSING BLOCK OF IMAGES.

Function	Parameter	6DIV value	26DIV value	27DIV value
<code>count_dapi_nuclei</code>	<code>clahe</code>	True	True	True
<code>count_dapi_nuclei</code>	<code>sharpen</code>	True	True	True
<code>count_dapi_nuclei</code>	<code>tophat</code>	True	True	True
<code>count_dapi_nuclei</code>	<code>low_intensity_tresh</code>	20	20	20
<code>count_dapi_nuclei</code>	<code>min_area_nuc</code>	40	40	40
<code>count_dapi_nuclei</code>	<code>max_area_nuc</code>	1500	1300	2000
<code>count_dapi_nuclei</code>	<code>kernel</code>	[[0,-1,0],[-1,7,-1],[0,-1,0]]	[[0,-1,0],[-1,7,-1],[0,-1,0]]	[[0,-1,0],[-1,7,-1],[0,-1,0]]
<code>find_blue_zones</code>	<code>sharpen</code>	True	True	True
<code>find_blue_zones</code>	<code>kernel</code>	[[0,-1,0],[-1,5,4,-1],[0,-1,0]]	[[0,-1,0],[-1,5,4,-1],[0,-1,0]]	[[0,-1,0],[-1,5,4,-1],[0,-1,0]]
<code>find_blue_zones</code>	<code>min_blue_area</code>	250	200	300
<code>find_blue_zones</code>	<code>hue_min</code>	40	40	40
<code>find_blue_zones</code>	<code>hue_max</code>	140	140	140
<code>find_blue_zones</code>	<code>sat_min</code>	10	10	10
<code>find_blue_zones</code>	<code>val_min</code>	15	15	15
<code>find_blue_zones</code>	<code>blue_dom_thresh</code>	1	1	1
<code>find_blue_zones</code>	<code>morph</code>	open(9x9)x2 + close(9x9)x1	open(9x9)x2 + close(9x9)x1	open(9x9)x2 + close(9x9)x1
<code>count_bgal_nuclei</code>	<code>near_radius</code>	40	42	42
<code>count_bgal_nuclei</code>	<code>ignore_aggl</code>	True	True	False
<code>count_bgal_nuclei</code>	<code>agg_tol</code>	0.5	0.5	None

E. Nuclei Segmentation

Contrast enhancement (CLAHE) and sharpening are applied to improve nuclear boundaries, followed by morphological opening. Grayscale Otsu-thresholding yields a binary mask; external contours are extracted and classified by area. Single-nucleus centroids are computed from moments and used in downstream spatial tests.

F. Senescent-Cell Extraction

A hybrid color rule is applied to the β -gal image to increase robustness against illumination and shade:

$$D_{\text{abs}} = \max\left(0, B - \frac{R+G}{2}\right), \quad (1)$$

$$D_{\text{rel}} = \frac{D_{\text{abs}}}{V + \epsilon} \cdot 255. \quad (2)$$

Pixels are selected if they pass both HSV hue windowing ($H \in [40, 140]$, $S \geq 10$, $V \geq 15$) and $D_{\text{rel}} > \text{blue_dom_thresh}$. After morphological open/close, external contours define candidate senescent regions. A nucleus is counted senescent if its centroid lies inside the contour; if none, the nearest single nucleus within `near_radius` is counted.

III. RESULTS

A. Comparison Across DIV (Days In Vitro)

Figure 1 summarizes the automated and human measurements grouped by DIV for three metrics: total nuclei count (`dapi_n`), senescent-positive count (`n_bgal`), and senescent percentage (`senescent_perc`).

For `dapi_n`, the automated method produced significantly higher counts at DIV26 ($p = 0.011$) but showed no significant difference at DIV27 ($p = 0.230$). For `n_bgal`, no statistically significant differences were observed at either DIV ($p > 0.05$). However, `senescent_perc` exhibited significant differences at both DIV26 ($p = 0.001$) and DIV27 ($p = 0.048$), indicating that while β -gal-positive cell counts may align, the disagreement in total cell numbers propagates into percentage estimates.

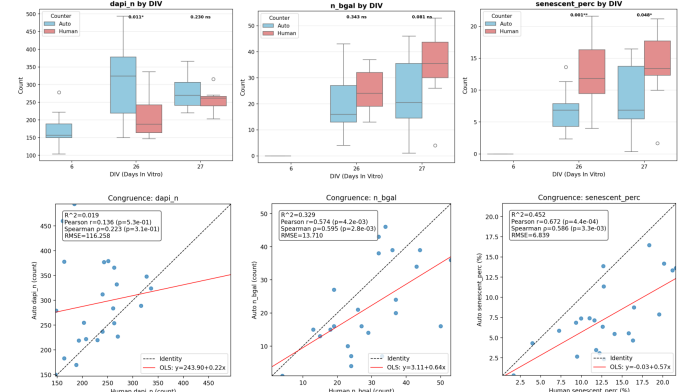


Fig. 1. Automated vs. human results. Paired analysis considering different conditions according to Days *in Vitro*. Statistical t-test was performed for each pair of measurements to assess congruence. Significance levels: *** $p < 0.001$, ** $p < 0.01$, * $p < 0.05$, ns=not significant

B. Congruence Between Automated and Human Counts

Scatter plots for dapi_n, n_bgal, and senescent_perc demonstrated mixed levels of agreement. Nuclei counts exhibited minimal linear correspondence ($R^2 = 0.019$; Pearson $r = 0.136$) with high error (RMSE = 116.3). In contrast, β -gal counts showed moderate congruence ($R^2 = 0.329$; Pearson $r = 0.574$, $p < 0.01$), and the senescence percentage demonstrated the strongest association ($R^2 = 0.452$; Pearson $r = 0.672$, $p < 0.001$). These results suggest that the β -gal segmentation module aligns more closely with human evaluation than nuclei counting. An example of processed images can be observed in Figure 2

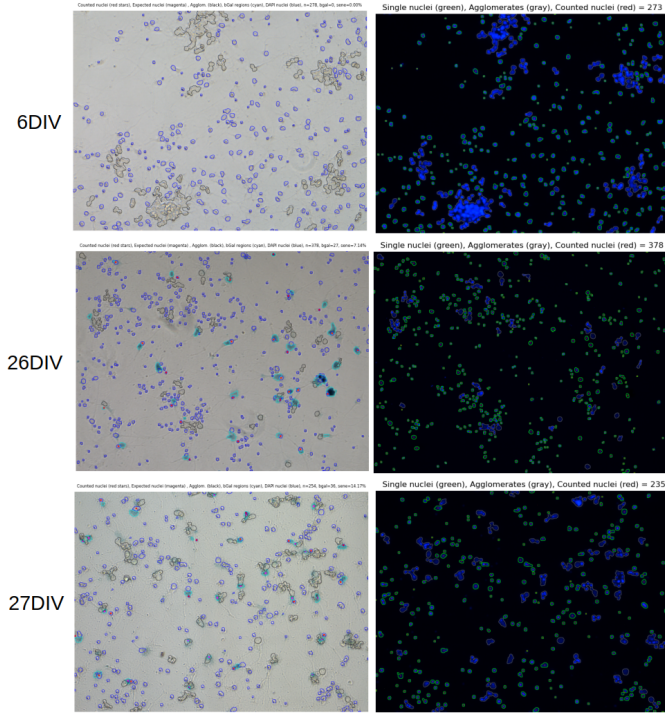


Fig. 2. Processed images with drawn contours of single nuclei (blue in left pictures and green in right pictures), nuclei agglomerations (gray), and β -gal zones (cyan). Each red point indicates which nuclei were included in the count (n_bgal in left pictures, n_dapi in right pictures)

C. Bland-Altman Analysis

Bland-Altman plots (see on Figure 3 revealed systematic biases across all three metrics. Automated dapi_n counts were consistently higher than human counts (mean bias = +64.9 nuclei), with wide limits of agreement ($[-128, +258]$), indicating substantial overestimation. For n_bgal, the mean bias was smaller (+7.43), but variability remained non-negligible ($[-30.5, +15.6]$). Senescent_perc showed a negative bias (mean = -5.67%), confirming that the pipeline tends to underestimate senescence frequency despite relative agreement in senescent-positive counts.

D. Residual Distribution and Normality

Residual histograms and Q-Q plots demonstrated non-normal, skewed error distribution for dapi_n, driven primarily

by systematic over counting. Conversely, n_bgal and senescent_perc residuals were more symmetrically distributed, with Q-Q plots indicating approximate normality. This pattern is consistent with the observation that β -gal segmentation performs more reliably than nuclei segmentation, and that errors in dapi_n directly impact senescence percentage.

The system consistently underestimated senescence relative to manual scoring. Bland-Altman and residual analyses confirm negative bias and limited agreement spread ($\pm 9.3\%$ bounds). Nuclei counting was generally accurate, with error driven primarily by staining-intensity variation in β -gal segmentation.

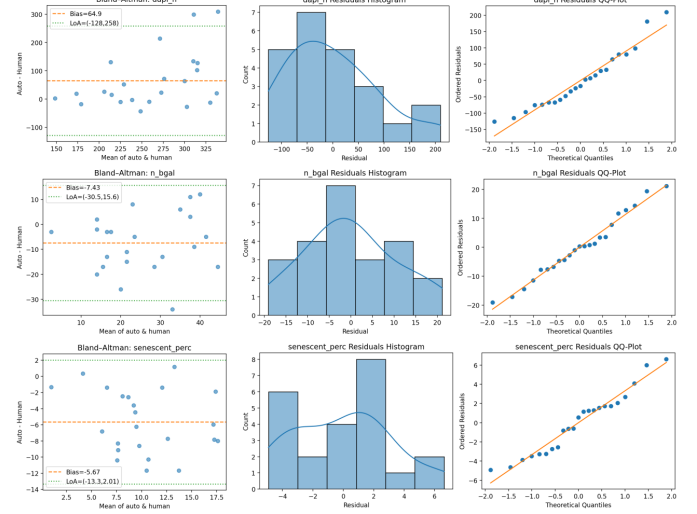


Fig. 3. Bland-Altman plots, residual histogram and residual QQ-plot comparing automated and human scoring.

IV. DISCUSSION

While the algorithm performs consistently, under-counting senescent cells indicates insufficient sensitivity to weak or diffuse β -gal signal. On the contrary overestimation of total cells may be not be entirely by detection of spurious nuclei objects but instead greater sensitivity to detect nuclei than human annotation. Overall the final senescent percentage significantly different to the validation results (p-val of 0.001 and 0.048 for 26 and 27 DIV respectively) but still poses a positive correlation with human validation results (obtaining an $R^2 = 0.452$ and Pearson correlation of 0.672) which highlights the complexity of the task to homologate human and automatic results but offers a hope for future improvement. We expect that machine-learning segmentation could reduce bias while a hyper-parameter optimization could further improve our results.

V. CONCLUSION

We presented an automated pipeline for senescent-cell quantification using DAPI nuclear staining and SA- β -gal chromogenic imaging. The pipeline for automated quantification integrates Otsu thresholding nuclei separation and a hybrid

color-segmentation strategy that combines hue filtering, absolute blue dominance, and illumination-normalized relative blue dominance. These pipeline tries to be more robust over simple threshold-based approaches, particularly in images with uneven staining, variable illumination, and specially overlapping cells.

However, the nuclei segmentation module presents limitations. Otsu separation of DAPI stained nuclei tended to overcount nuclei, producing a large positive bias (+64.9) and poor linear correspondence with human evaluation ($R^2 = 0.019$). Because senescence percentage depends on both β -gal and DAPI counts, inaccuracies in total nuclei estimation propagated into final senescence estimates, resulting in systematic underestimation (-5.67% mean bias).

In conclusion, the automated pipeline represents a substantial step and improvement over simple threshold-based methods but still falls short of human-level accuracy in its current form. The future work suggested is refinement of the nuclei segmentation component to create a reliable automated senescence quantification.

FUTURE WORK

Suggestions to improve the automated senescence quantification:

- **Advanced Color Normalization**
- **Adaptive and Local Thresholding**
- **Machine Learning and Deep Segmentation Models**
- **Per-Image Calibration**
- **Multichannel or Multimodal Imaging**
- **Large-Scale Evaluation**
- **End-to-End Automation**

Future work incorporating these suggestions are expected to significantly reduce bias, improve agreement with human scoring, and ultimately enable scalable, reproducible, and reliable quantification of cellular senescence in experimental pipelines.

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VI. CODE AVAILABILITY

Code is available at <https://github.com/raqcoss/senescence-counter>

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